

THE PROBLEM

01

WASTE GENERATION

- IIT(BHU) generates a substantial amount of waste daily, stemming from various sources such as residential areas, academic buildings, hostels, and dining facilities.
- The diverse nature of this waste complicates effective management and disposal.

Table 1: Major point sources and types of MSW in the IIT (BHU) campus

S.no	Sources	A major type of components
1.	Hostels	Food, paper, plastic, thermacol, fruit peels, clothes, Metals, and dust
2.	Mess	Food waste and other waste
3.	Canteens	Plastics, eggs shells and vegetable peelings, food waste
4.	Fruit shops	Fruit peelings, plastic glasses, mug, etc.
5.	Vishwanath Temple	Flowers, leaves, plastic and earthen bowls
6.	Residential quarters	Food, paper, plastic, clothes, vegetables and fruit peels, clothes, metals, and dust

02

CURRENT DISPOSAL METHODS

- The existing waste management system at IIT(BHU) primarily involves collection and transporting waste to landfills.
- There is no facility of treatment and recycling process within the campus. Hence, all the waste collected is transported to the dumping ground

03

ENVIRONMENTAL IMPACT

- The uncontrolled dumping of waste in landfills contributes to soil and groundwater contamination.
- If it is not well managed and disposed off scientifically, it will pose risks to human health, other living organisms and environmental effects (CO₂ and CH₄ emissions).
- The accumulation of unmanaged waste poses health risks to the campus community, attracting pests and creating breeding grounds for disease-causing organisms.



Landfill in IIT-BHU

What We Concluded From Waste

Biodegradable

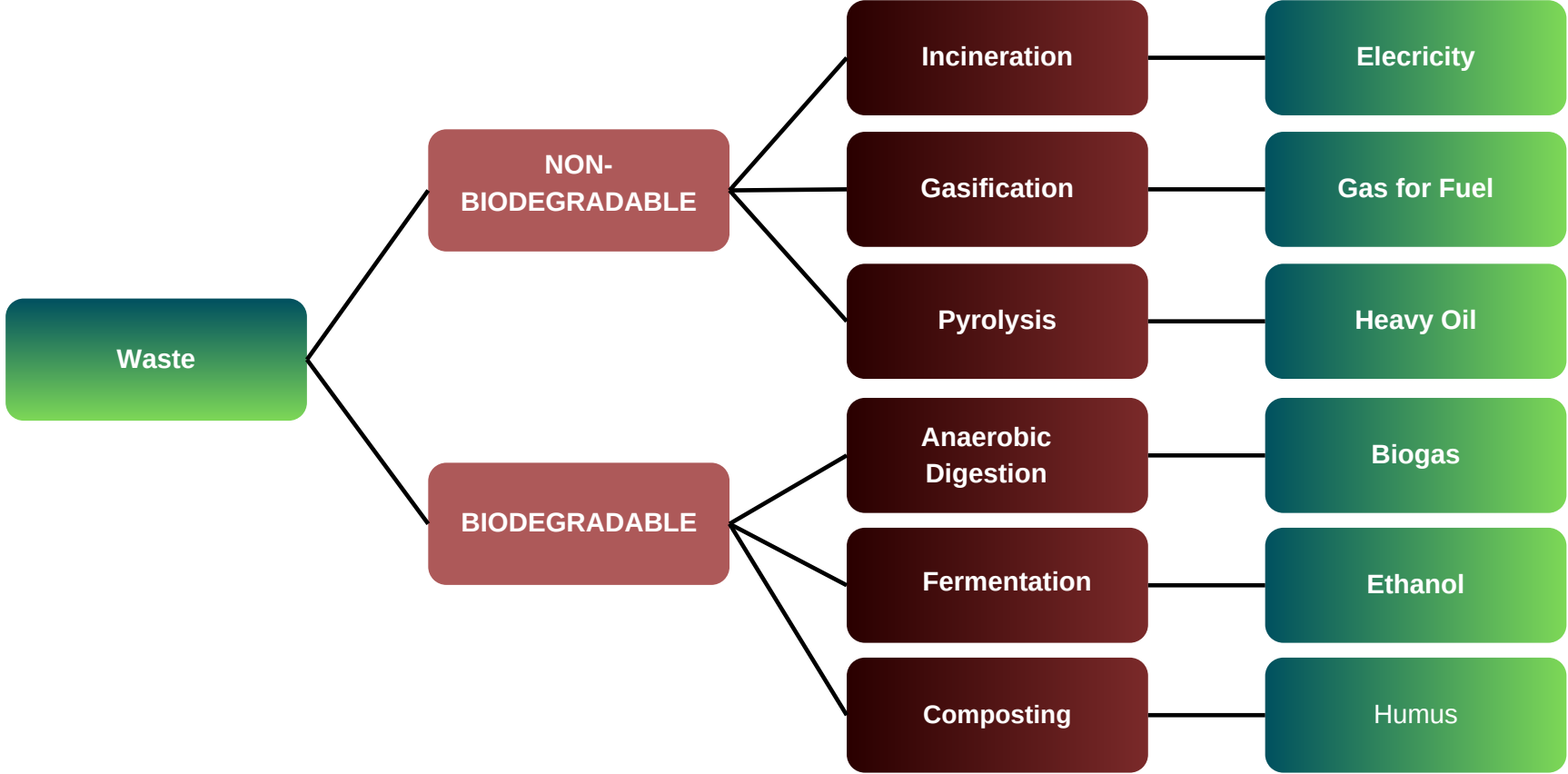
Biodegradable refers to materials that can break down and decompose naturally through the action of microorganisms like bacteria, fungi, and other biological processes. These materials, such as food waste, paper, and plant matter, are typically converted into natural substances like water, carbon dioxide, and organic matter.

Non-Biodegradable

Non-biodegradable, on the other hand, describes materials that do not break down easily or decompose naturally. These include plastics, metals, and synthetic materials which can persist in the environment for hundreds or even thousands of years, often contributing to pollution and environmental degradation

Table shows a solid waste generation from the IIT (BHU) campus.

Components	Min(kg)	Max(kg)	Avg.(kg)
Biodegradable	351	525	409
Non-Biodegradable	312	548	411
Total	663	1073	820



Waste generation rates work out to be approximately 409 kg/day biodegradable and **411 kg/day** of nonbiodegradable materials i.e., **820 kg /day** MSW is generated from the IIT (BHU) campus in a day. Hostel mess activities generate about **1060 kg/day** that do not enter the collection system of MSW in the campus. This is because most of the food waste generated by the hostel mess is used as cattle feed.

NON-BIODEGRADABLE

01

INCINERATION (COMBUSTION)

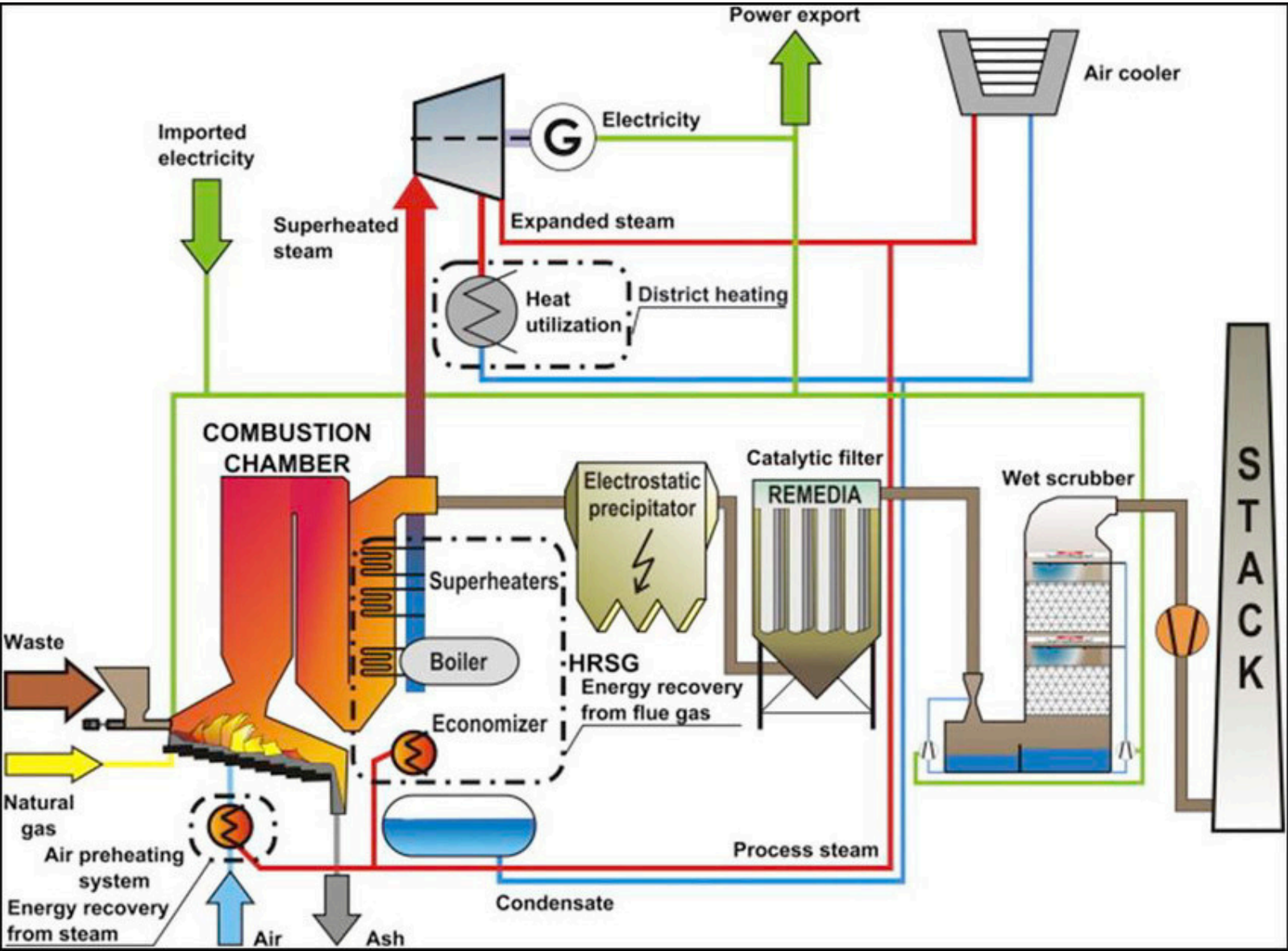
Principle: Incineration involves burning waste at very high temperatures (often exceeding 850°C) to convert it into heat, ash, flue gas, and other byproducts. The heat generated is used to produce steam, which drives turbines to generate electricity or provide district heating.

Challenges:

- Air pollution from the release of dioxins, furans, and particulate matter.
- Expensive emission control technologies are required.
- High capital and operational costs.

Desirable range of important waste parameters for technical viability of energy recovery:

Waste Treatment Method: Incineration	Basic principle: Decomposition of organic matter by action of heat.	Important Waste Parameters		Desirable Range
		Moisture content		< 45 %
		Organic/ Volatile matter		> 40 %
		Fixed Carbon		< 15 %
		Total Inerts		< 35 %
		Calorific Value (Net Value)	Calorific Value	>1200 k-cal/kg



NON-BIODEGRADABLE

02

GASIFICATION

Principle: Gasification involves heating waste in a controlled, low-oxygen environment to break it down into a mixture of gases, primarily hydrogen (H₂) and carbon monoxide (CO), known as syngas. Syngas can be used to produce electricity, and heat, or as a feedstock for producing chemicals or fuels.

Process:

- Waste is fed into a gasifier exposed to high temperatures (800-1000°C) with limited oxygen.
- This breaks down the waste into syngas and solid residue (char).
- The syngas are cleaned and can be used to generate power in turbines or internal combustion engines.

Advantages:

- More efficient than traditional combustion for certain waste streams.
- Syngas can be used for electricity or converted into liquid fuels like methanol or synthetic diesel.
- Produces less harmful emissions than incineration.

Drying of Biomass: $\text{H}_2\text{O (liquid)} \rightarrow \text{H}_2\text{O (gas)}$

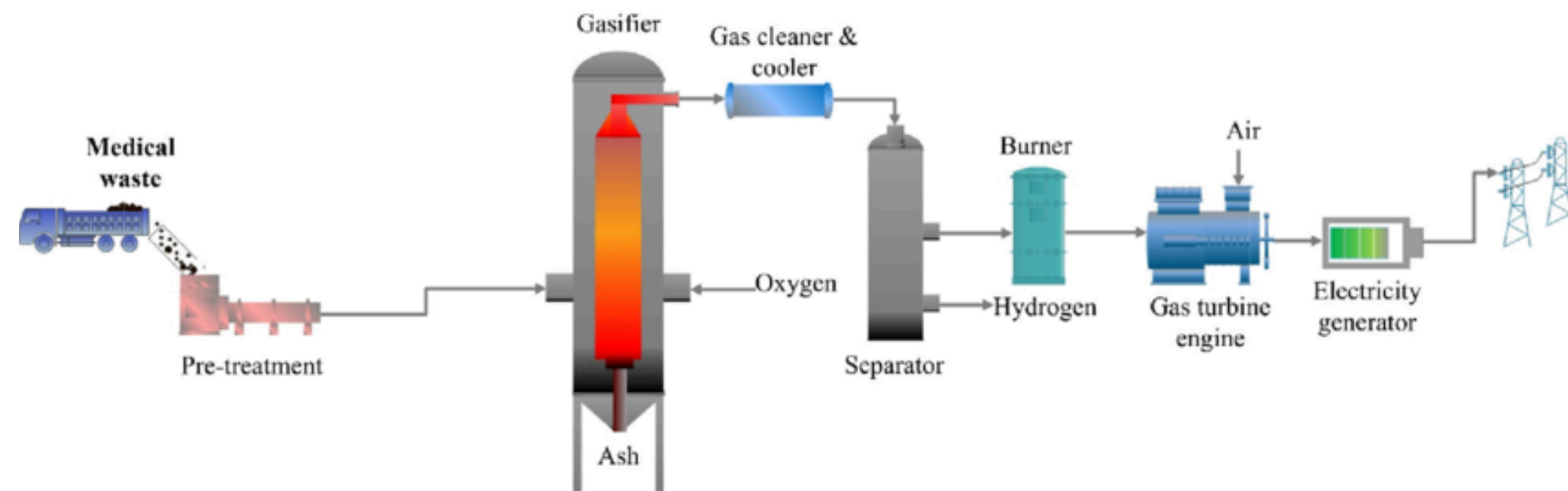
Pyrolysis: $\text{C}_n\text{H}_m \xrightarrow{\text{Heat}} \text{Char} + \text{Tar} + \text{Volatile gases}$

Carbon Reactions:

- Carbon to Carbon Dioxide: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- Carbon to Carbon Monoxide: $\text{C} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}$

Water Gas Reaction: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$

Overall Gasification Reaction: $\text{C} + \text{H}_2\text{O} + \text{O}_2 \rightarrow \text{CO} + \text{H}_2 + \text{CO}_2$



Biodegradable Waste Process

01

Anaerobic Digestion (AD)

Principle: This is a biological process where microorganisms break down organic matter (e.g., food waste, agricultural waste, sewage) in the absence of oxygen, producing biogas (methane + CO₂) and digestate (a nutrient-rich slurry).

Process:

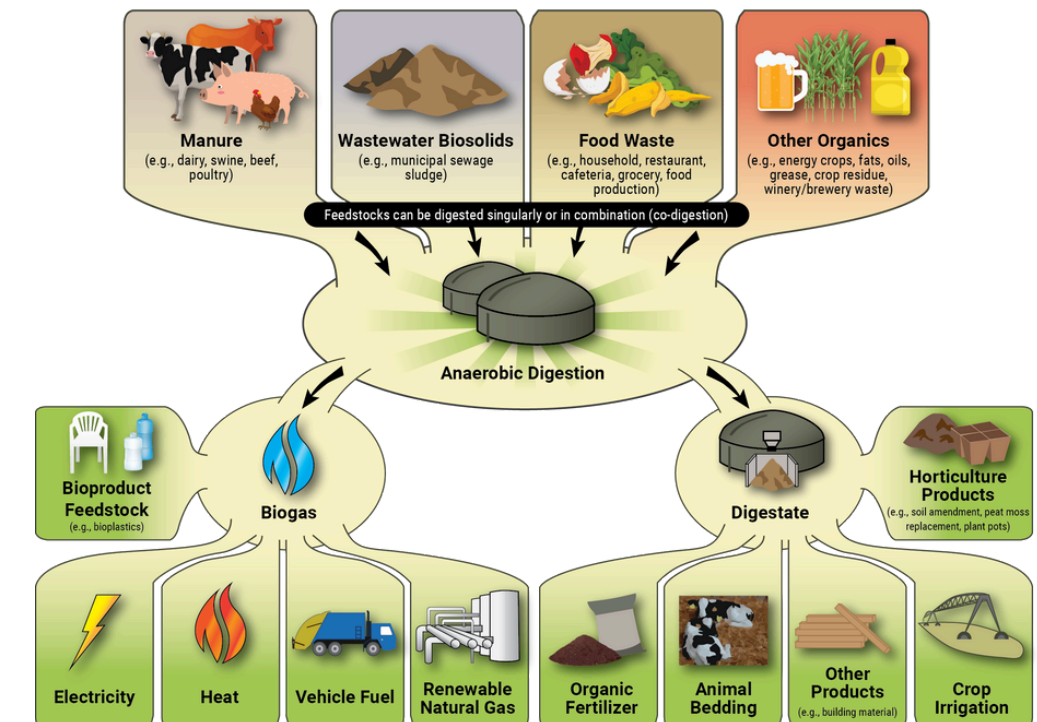
- Waste is fed into a gasifier exposed to high temperatures (**800-1000°C**) with limited oxygen.
- This breaks down the waste into **biogas** and **solid residue (char)**.
 1. **Hydrolysis:** Complex organic polymers (like carbohydrates, proteins, and fats) are broken down into simpler molecules (like sugars, amino acids, and fatty acids) by hydrolytic bacteria.
 2. **Acidogenesis:** Acid-forming bacteria convert the simpler molecules from hydrolysis into organic acids (like acetic acid, propionic acid, and butyric acid), as well as carbon dioxide and hydrogen.
 3. **Acetogenesis:** Acetogenic bacteria convert the organic acids from acidogenesis into acetic acid, carbon dioxide, and hydrogen.

Applications:

- **Energy Production:** The biogas can be used to generate electricity, heat, or transportation fuels, such as biomethane
- **Agriculture:** Digests livestock manure and food waste to produce biogas and fertilizer.

Challenges:

- High capital and maintenance costs.
- Complex gas cleanup technologies are needed to remove contaminants.
- Best suited for homogeneous and dry waste, like wood chips or plastics.



Biodegradable Waste Process

02

FERMENTATION

Principle:

- Fermentation is an anaerobic process where microorganisms convert organic compounds (like sugars) into acids, alcohols, or gases. The products vary based on the microorganisms and substrates used.

Process:

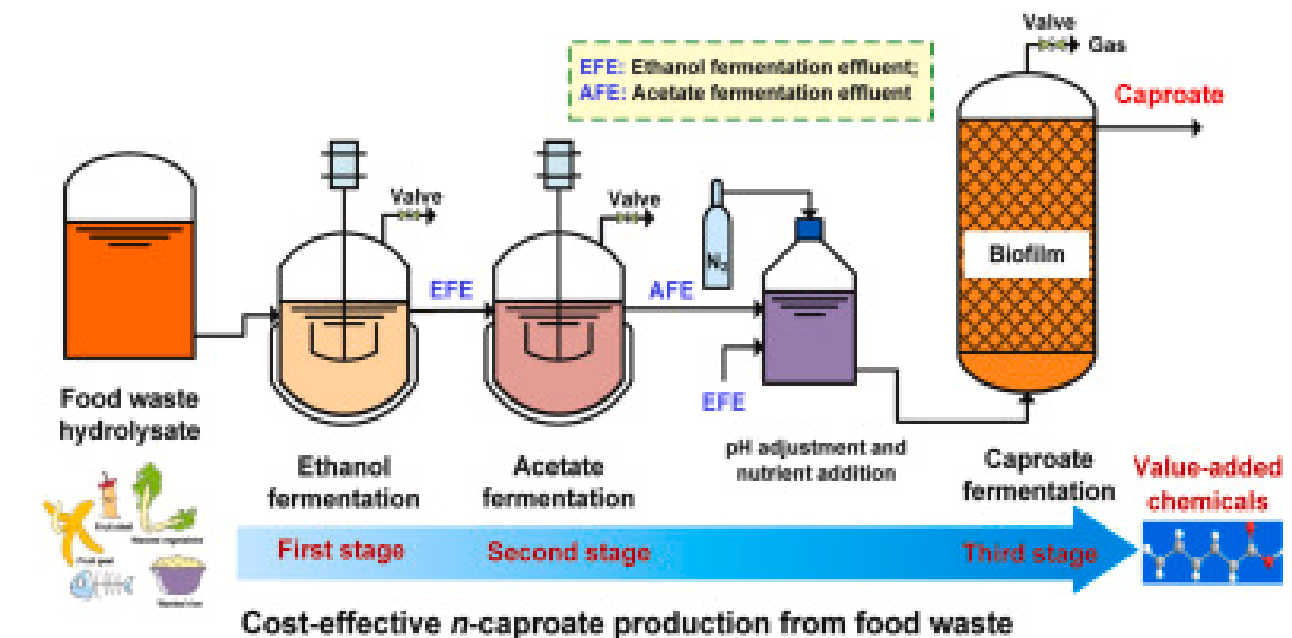
- Organic material (sugars) is used as a substrate.
- Microorganisms (yeast, bacteria) metabolize the substrate in oxygen-free conditions.
- Products like ethanol, lactic acid, or gases are formed.

Applications:

- Food: Used to make yogurt, cheese, bread, and alcoholic beverages.
- Biofuels: Produces bioethanol from plant sugars.
- Waste Treatment: Breaks down organic waste to produce biogas and other byproducts.

Challenges:

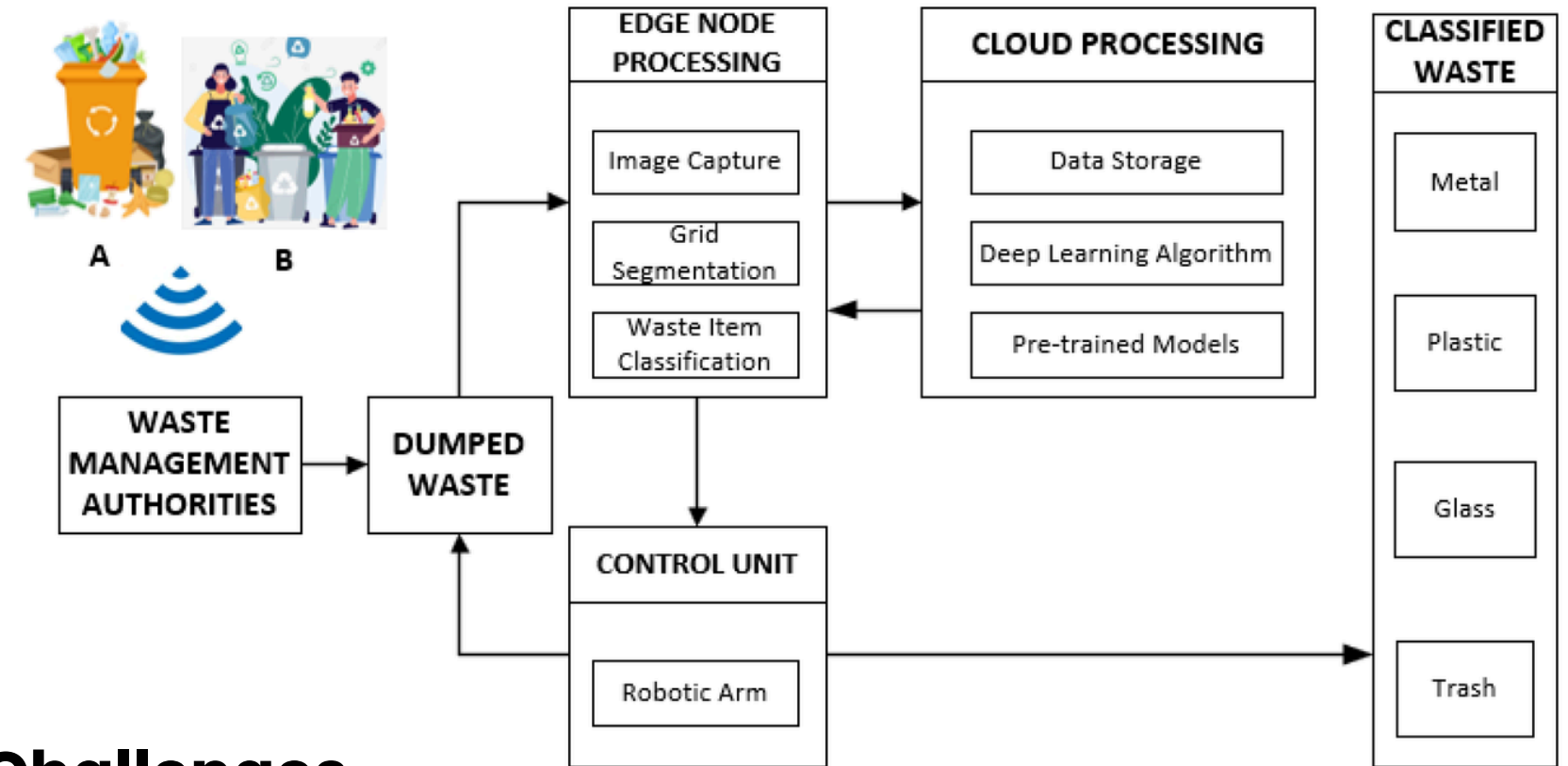
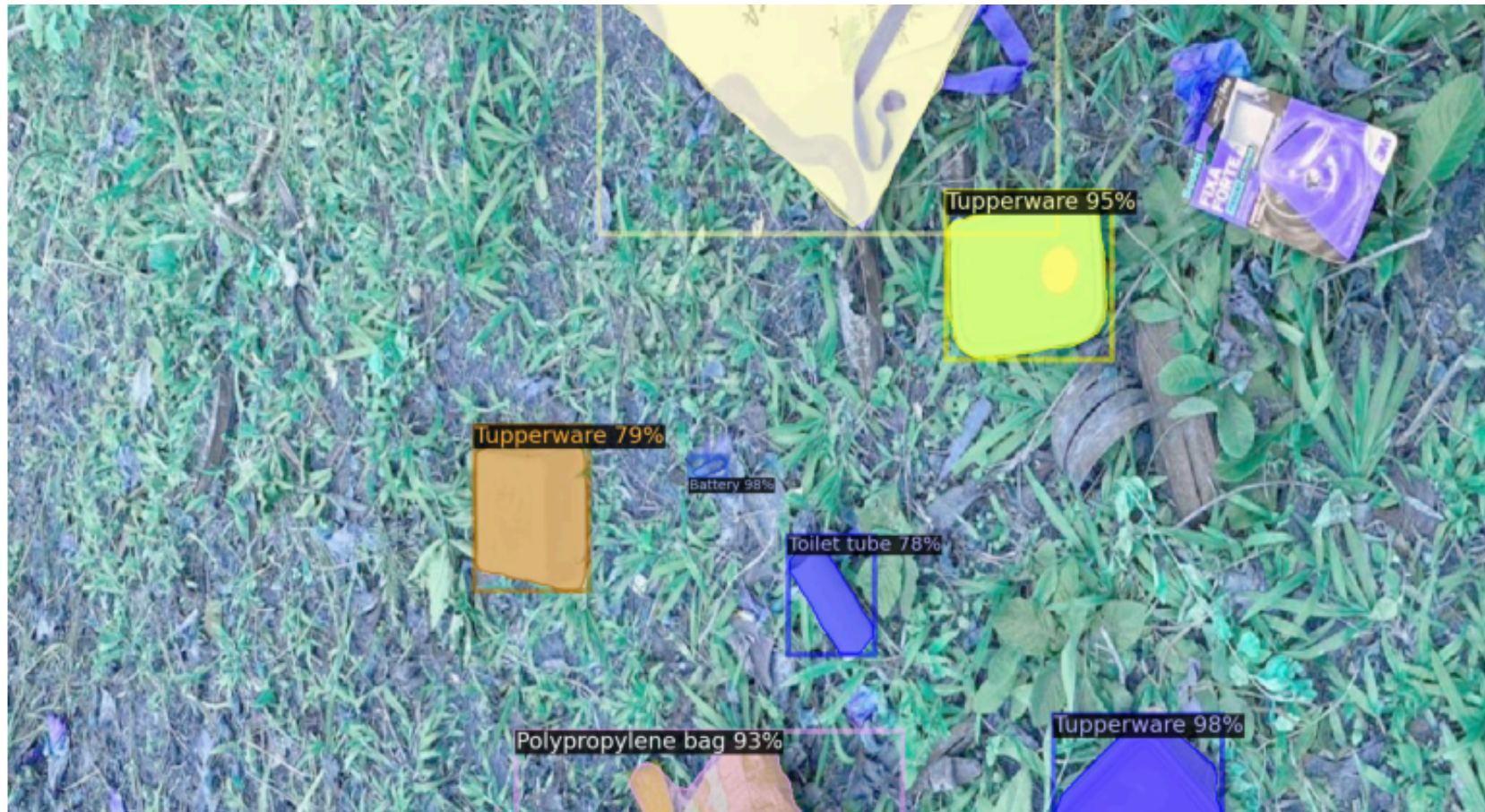
- Substrate availability and cost.
- Process efficiency and yield.
- Risk of contamination.
- Energy demands and byproduct management.



WASTE SEGREGATION

Important points of model

- Implemented **CNN model** Based on **Resnet50**
- Acheived final train accuracy = **91.81%** ,
Validation accuracy = **89.37%**
- Acheived overall **F1 score of 0.16** and support of **2333**
- Essential libraries like **TensorFlow, Keras, NumPy, Pandas,** and **Matplotlib** are imported for model creation and data manipulation.



Challenges

- Without access to high-end hardware (e.g., those relying on CPUs or lower-end GPUs) will experience slow training time.
- Indian moms pack food in black polythene bags, making it challenging for the model to predict accurately.
- Certain waste items, such as plastic and metal, may look similar in certain conditions (e.g., reflective surfaces, translucent materials), leading to misclassification.
- Some waste types, such as plastic, are more common than others like metal or glass, leading to an unbalanced dataset that biases the model.

Preliminary Results

Anaeoribic digestion

- Assumed average biogas yield is **0.5 m³/kg** of biodegradable waste, influenced by composition, temperature, retention time, and the anaerobic digestion process used.
- From **409 kg** of biodegradable waste, using anaerobic digestion, you could generate approximately **250 to 375 kWh** of energy, depending on the biogas yield.

Fermentation

- Assumed waste is mainly organic (e.g., food, agricultural residues) for anaerobic digestion, yielding 0.5 m³ of biogas/kg, varying with material type and fermentation conditions.
- Biogas Yield: **100 to 200 m³/ton** and Energy Content: **20 to 25 MJ/m³** and Total Energy: **~227.2 to 567.6 kWh** from **409 kg** of biodegradable waste

Incineration

- Assumed average calorific value is **8 MJ/kg** for mixed non-biodegradable waste, with **85%** incineration efficiency for heat-to-electricity conversion, varying based on waste composition.
- Total Energy from **411 kg** of Non-Biodegradable Waste via Incineration is about **~1,142 kWh to 2,283 kWh**.

Pyrolysis

- It is assumed that the pyrolysis of **411 kg** of non-biodegradable waste yields approximately **50% bio-oil, 10% gas, and 20% char**, reflecting typical values for such waste types.
- Total Energy is about **~1,713 kWh to 3,650 kWh** from **411 kg** of non-biodegradable waste via pyrolysis.

CHALLENGES

- **Segregation and Contamination:**

A lack of effective waste segregation at source is a major obstacle, as both biodegradable and non-biodegradable wastes often end up mixed.

- **Toxic and Hazardous Materials**

Some inorganic waste can produce toxic by-products during chemical processing, such as heavy metals, dioxins, or other hazardous chemicals.

- **Technology Selection**

Choosing the right WtE technology (incineration, gasification, anaerobic digestion, etc.) that fits IIT BHU's specific waste profile is critical. A mismatch between waste type and technology can result in low efficiency and operational difficulties.

- **Variability in Waste Appearance**

A model trained on clean and well-defined waste samples may struggle to detect waste in messy or mixed environments, reducing its utility on the campus where waste might not always be well segregated or visually clear.

- **Visual Similarity of Waste Types**

The model may misclassify similar-looking materials, reducing its overall accuracy. Misclassifications can hinder waste segregation, affecting the efficiency of the WtE conversion process.

